TEC-C ARCHITECTURE & EXECUTIVE PLAIN-ENGLISH GUIDE

How Tec-C Works: Complete System & Flow

Autonomous AI Capital Management with Mathematical Blockchain Leashes

> **The Core Metaphor:** Imagine hiring a brilliant Wall Street trader to manage your money, but instead of giving them blank checks, you give them a smart corporate card on a **mathematical leash**. Within your daily budget ($500/day) and on approved blue-chip protocols (Aave, Uniswap), the AI trades autonomously 24/7. If it finds a massive $5,000 opportunity or wants to touch a new asset, it cannot spend a single cent without your **FaceID / TouchID biometric approval**.

01 The Three Foundation Pillars of Tec-C

1. THE AI BRAIN

The Intelligent Driver. Listens to your plain English commands, monitors live market weather (fear/greed, liquidation risks), scans for yields, and calculates stablecoin arbitrage.

2. THE SMART VAULT ("THE TETHER")

The Blockchain Leash. A smart contract wallet (ERC-4337) that enforces mathematical daily spending limits, protocol whitelists, and risk tiers before any trade can touch the blockchain.

3. MOBILE VIRTUAL SESSION

The Master Remote Key. Runs on your local laptop, but lets you monitor trades, receive alerts, and confirm high-value trades via FaceID on your phone from anywhere in the world.

02 The Big Problem Tec-C Solves

Before Tec-C, crypto investors faced two broken extremes:

Feature / Risk	Traditional Trading Bots	Standard Hardware Wallet	Tec-C Autonomous Vault
Autonomous 24/7 Trading	✓ Full autonomy	✕ Zero (Manual click for everything)	✓ Full autonomy within daily limits
Catastrophic Hack Risk	✕ High (API keys can drain 100%)	✓ Low (Keys kept offline)	✓ Zero (Capped by strict on-chain leash)
Seed Phrase Vulnerability	✕ Vulnerable to leaks & phishing	✕ 12 words written on paper	✓ None (Hardware FaceID / Passkeys)
Stablecoin Crash Protection	✕ Blind to depegs	✕ Too slow (You must be awake)	✓ PegGuard auto-swaps before loss
Remote Phone Monitoring	✕ Requires custodial cloud servers	✕ Requires carrying physical USB dongle	✓ Private encrypted phone session

03 The Mathematical Leash: How the Vault Protects You

Every transaction the AI initiates must pass through the **TetherGuardHook** smart contract. It enforces three unbreakable rules:

1. Daily Velocity Limit: The AI can only spend up to your daily budget (e.g. $500/day). The allowance automatically resets every 24 hours.

2. Vetted Whitelist Only: The AI can never send funds to unknown or malicious addresses. It can only talk to audited blue-chip contracts.

3. 5-Minute Biometric Window: High-value trades over the daily limit unlock a temporary 5-minute window only when you authenticate with FaceID.

04 Step-by-Step Flow: How a User Command Becomes a Trade

Here is the exact journey of what happens behind the scenes from the moment you speak or type a command until profits hit your vault:

1 **User Command in Plain English** `Input`

You type or speak naturally into the app: "*Find the best yield for 5,000 USDC on blue-chip pools, but avoid Curve.*" You don't need to know complex DeFi routing or smart contract methods.

2 **Security Firewall & Privacy Sieve** `Safety`

Before your words reach any AI model, a local security firewall verifies that the prompt has no malicious prompt-injection attacks. It converts your request into a strict mathematical **Mandate**: `Asset: USDC | Action: YIELD | Amount: 5000 | Avoid: Curve`.

3 **Sentinel Market Weather Radar** `Intelligence`

The Brain scans live market indicators: **CMC Fear & Greed Index** (market mood), **Coinglass** (whale leverage & liquidation dangers), and **Real-time News** (exploits/breaking events). If a flash crash or market panic is underway, it automatically tightens risk limits.

4 **Opportunity Search & PegGuard Scanning** `Discovery`

The Brain scans whitelisted protocols (Aave, Compound, Uniswap, Spark, Morpho) for top yields. Concurrently, **PegGuard** inspects 25+ stablecoins across DEXs for depeg risks and price-difference arbitrage opportunities.

5 **Hard Risk Filters & Asymmetric Scoring** `Risk Math`

Every candidate opportunity must pass strict math tests: pool depth, smart contract audit rating, and depeg probability. The math model uses *loss-averse scoring* (downside risk is penalized 2.5x more than upside gain). Surviving candidates are ranked as **Good, Better, and Best**.

6 **Bounded AI Decision & Plain-English Explanation** `Synthesis`

The LLM evaluates the pre-vetted options. **The Golden Safety Rule:** The LLM can choose among candidates or reduce trade size, but it can *never* add unapproved venues or spend more money. It generates a clear rationale: "*Selected Aave v3 USDC at 9.2% APY with $45M reserve safety. Avoided Curve per your instructions.*"

7 **The Tether Gatekeeper & On-Chain Execution** `Execution`

The trade is dispatched to your Smart Vault:

• **Within Daily Limit ($300 trade vs $500 limit):** Executes automatically on-chain. Zero friction.

• **Breakout Trade ($5,000 trade vs $500 limit):** Pauses and sends a prompt to your mobile phone. You scan FaceID, opening a 5-minute window that completes the trade.

05 PegGuard: Stablecoin Radar & Arbitrage

How PegGuard Protects & Grows Your Balance:

Stablecoins are supposed to stay at exactly $1.000. But during turbulence, they fluctuate:

- **Crash Defense:** If a coin you hold slips below $0.985 (like UST in 2022), PegGuard immediately detects the anomaly and swaps your balance into safe blue-chip collateral before you lose money.
- **Risk-Free Arbitrage:** If DAI drops temporarily to $0.993 on one exchange while trading at $1.001 on another, PegGuard captures the price gap automatically, earning free yield while keeping your money in dollars.

06 Real-World Scenarios: Tec-C in Action

Here is how Tec-C behaves in everyday life under different market conditions:

Scenario A: The Silent Daily Yield Earner

100% Autonomous

Situation: You have $10,000 sitting idle in your Tec-C vault with a daily spending allowance of $1,000.

AI Discovery: The AI Brain spots that Compound v3 is offering a flash 11.2% lending yield on USDC.

What Happens: The AI allocates $800 into the pool. Because $800 is under your $1,000 daily leash, the smart contract executes the deposit silently on-chain.

User Experience: You wake up in the morning, check your dashboard, and see that you earned extra yield while you were asleep—without having to approve a thing.

Scenario B: The High-Value Breakout Opportunity

Biometric FaceID Required

Situation: Your daily limit is $1,000. The AI Brain spots a massive arbitrage opportunity that requires a $4,500 swap to capture a $120 profit in 30 seconds.

What Happens: The smart contract detects that $4,500 exceeds your $1,000 leash. It blocks autonomous execution and sends an urgent push alert to your phone.

User Action: Your phone buzzes: "*High-Profit Arbitrage Opportunity: $4,500 swap to gain $120. Tap to approve.*"

Result: You glance at your screen, verify the profit, scan FaceID, and the 5-minute biometric window opens. The trade finishes instantly.

Scenario C: PegGuard Defense in a Market Panic

Crisis Circuit-Breaker

Situation: You hold 6,000 units of a synthetic stablecoin. Breaking news hits that the protocol's backing reserves are under stress.

Sentinel Alert: The market radar detects social panic and rising liquidation cascades; the stablecoin price slips to $0.982.

PegGuard Action: Within seconds, PegGuard activates its automated defense protocol and swaps the 6,000 synthetic coins into native USDC at $0.981.

Result: An hour later, the troubled stablecoin crashes to $0.40. While other investors lost 60% of their money, your vault protected 98.1% of your capital.

Scenario D: Away From Desk (Mobile Virtual Session)

Remote Phone Access

Situation: Tec-C is running securely on your home laptop. You are traveling or at a cafe with just your phone.

Setup: Before leaving, you clicked "*Create Virtual Session*". Your laptop created an encrypted, private tunnel (e.g. `tec-c-session-8f3a9b.trycloudflare.com`).

Experience: You open the link on your mobile phone browser and scan your phone's fingerprint. The session stays connected continuously until you disconnect. You monitor trades, adjust risk, and approve transactions on the go.

Scenario E: The Lost Phone Emergency

Zero Seed Phrase Risk

Situation: You accidentally drop your iPhone in the ocean. How do you recover your money without a 12-word seed phrase?

Multi-Device Passkey: Because you linked both your iPhone and your MacBook TouchID during setup, you can immediately open your MacBook, log into your vault with TouchID, and revoke the lost iPhone's access key.

Emergency Guardian: Even if both devices were destroyed in a house fire, your pre-configured Recovery Guardian (a trusted hardware key or family multisig) can sign an authorization to attach a brand-new device. Your funds are never lost.